

Unit 1

Command Line Fundamentals



Windows Terminal & Bash / Linux

Navigate the file system and operate confidently from the shell.

Topics

File system navigation

Files & folders

Basic CLI operations

Relative Path vs Absolute Path

Commands

pwd
print working dir

ls / dir
list contents

cd
change directory

cd ..
up one level

cd ~
home directory

mkdir
make directory

touch / New-Item
create file

rm / Remove-Item
delete

cp / Copy-Item
cp original.txt backup.txt

cat
print file

nano
edit in terminal

tree
show dir tree

history
view command history

clear / cls
clear the screen

ren / mv
mv old_name.txt new_name.txt

Unit 2

Python Environment & Packages



Python setup, venvs & pip

Isolate projects with virtual environments and manage dependencies.

Topics

Python setup

Virtual environments

Package management

Commands

python -m venv venv
create virtual environment

deactivate
exit the venv

pip install
install a package

python -m pip freeze > requirements.txt
lock dependencies for reproducible installs

python --version
check Python version

pip list
list installed packages

pip uninstall
remove a package

Unit 3

Git Fundamentals



Version control with Git

Track changes, snapshot progress, and understand the repository lifecycle.

Topics

Version control	Repository lifecycle	Tracking changes
Working Directory	Staging Area	Local Repository

Commands

<code>git init</code> start a repo	<code>git status</code> working tree state	<code>git add</code> stage changes
<code>git commit</code> snapshot changes	<code>git log</code> history of commits	<code>git config --list</code> view configuration
<code>git config --global user.name</code>		<code>git config --global user.email</code>
<code>git --version</code> check Git version	<code>git log --stat</code> detailed history	<code>git restore</code> discard changes
<code>git restore --staged</code> unstage files	<code>git reset</code> move HEAD	<code>git revert</code> undo commits safely

Unit 4

GitHub & Remote Repositories



Hosting code in the cloud

Push local work to GitHub and sync with remotes.

Concepts

Repository	Owner	Clone	Push
Pull	Fork	SSH Auth	Remote (origin)
README.md		.gitignore	

Commands

<code>git clone</code> copy remote repo	<code>git push</code> send commits	<code>git pull</code> fetch + merge	<code>git fetch</code> download only
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Unit 5

Branching & Collaboration



Working together on GitHub

Branches, pull requests, and the collaboration surface.

Concepts

Branches

Merge

Pull Requests

Issues

Releases

Stars

Watch

Commands

git switch -c
create + switch to new branch

git branch
list branches

git merge
integrate branch history

git push -u origin
push and set upstream

git checkout
older way to switch branches (superseded by git switch)

Development Workflow

The end-to-end loop, step by step



From empty folder to pushed commits
Follow the arrows through each stage of a real project loop.

